# Group Software Development Project – Information Sheet

**Course:** NANO2142 – Introduction to Software Development
**Group Size:** 5 students per group
**Weighting:** 20 marks (out of 30 marks allocated for Continuous Assessment)

## 1. Purpose of the Project

This project gives students the opportunity to **apply programming and database concepts in a real-world context**. By completing this activity, students will learn to:

- Develop a **user-friendly UI/UX** using Qt or another suitable IDE.
- Design and implement a **database system** with SQL.
- Apply **C++ Object-Oriented Programming (OOP) concepts** (encapsulation, inheritance, polymorphism, etc.) in solving problems.
- Integrate these components into a **working software application**.

The goal is to strengthen students' ability to design practical, functional, and user-centered tools that support analytical and research-related tasks.

## 2. Project Scope

Each group will design and implement a software application that:

- Includes a **functional GUI** (buttons, menus, forms, etc.).
- Uses a **database backend** for storing and retrieving data.
- Demonstrates correct application of **C++ OOP concepts**.
- Addresses a practical scenario in Material and Nano Science Technology or a relatable student/academic context.

**Examples:**

- Laboratory Inventory Manager
- Student Attendance Tracker
- Research Data Logger
- Library Book Management System

## 3. Group Roles (recommended)

- **UI/UX Designer** – builds the interface.
- **Database Designer** – designs schema & SQL.
- **OOP Developer** – implements core logic.
- **Integrator** – connects DB, UI, and logic.
- **Documentation & Testing Lead** – prepares report and tests application.

## 4. Deliverables & Deadlines

1. **Proposal + Initial Presentation:**

    - Brief written proposal (2–3 pages) including project title, objectives, planned system components, and group member roles.
    - 5-minute oral presentation to outline the idea.

2. **Final Deliverables (Week 14-15):**

    - Working Software Application (executable with source code).
    - UI/UX Design Document (screenshots, flow diagrams).
    - Database Schema & SQL Scripts.
    - Final Report (10–12 pages) – including objectives, design, implementation, testing, and contributions of members.
    - Final Presentation (10 minutes) – live demonstration of the system.

## 5. Assessment Criteria (20 Marks)

| Criterion | Marks |
|---|---|
| UI/UX Design (clarity, usability, aesthetics) | 4 |
| Database Design & Functionality | 4 |
| C++ OOP Implementation | 6 |
| Integration & Functionality of Full System | 3 |
| Documentation & Presentation | 3 |
| **Total** | **20** |

**6. Evaluation Rubric**

| Criteria | Excellent (A) | Good (B) | Satisfactory (C) | Poor (D/F) | Marks |
|---|---|---|---|---|---|
| **UI/UX Design** | Highly intuitive, user-friendly, and visually appealing. Clear navigation and professional design. | Clear and usable, with minor issues in design or layout. | Functional but lacks clarity or good design practices. | Confusing, poorly designed, or incomplete. | 4 |
| **Database Design & Functionality** | Well-structured schema, correct normalization, efficient queries. Works smoothly with UI. | Schema and queries mostly correct, minor inefficiencies or gaps. | Database design functional but limited; some errors in queries. | Poor schema, incorrect or incomplete queries; DB not integrated. | 4 |
| **C++ OOP Implementation** | Strong use of OOP principles (encapsulation, inheritance, polymorphism). Code is clean, modular, and efficient. | OOP principles applied with some minor flaws. Code works but not fully optimized. | Limited application of OOP; concepts used superficially. | Incorrect or missing OOP implementation; poor code quality. | 6 |
| **Integration & Functionality** | All components (UI, DB, OOP) work seamlessly together. Software is stable and fully functional. | Components integrated with minor issues or bugs. | Integration incomplete; some parts work, others fail. | Components not integrated; software largely non-functional. | 3 |
| **Documentation & Presentation** | Clear, complete, and professional documentation. Excellent final presentation and demo. | Good documentation and presentation, some minor gaps. | Documentation and presentation meet minimum requirements. | Poor or missing documentation; weak or incomplete presentation. | 3 |
| | | **Total** | | | **20** |

**Notes for Students**

- Proposal + 5-min presentation (ungraded but mandatory to proceed).
- Final deliverables + demo due **Week 14**.
- Marks will be allocated based on the **quality of the final product and group contribution**.
- All group members must participate in the **final presentation**.